

Day 18

UVM: Universal Verification Language

Basics:

We are verifying an adder here, ie $sum = a + b$

UVM components needed:

- seq_item → holds a, b, expected_sum
- sequence → generates random a,b pairs
- driver → puts a,b onto adder input pins
- monitor → reads a,b,sum from adder output pins
- scoreboard → checks $sum == a + b$
- agent → packages driver+monitor+sequencer
- env → contains agent+scoreboard
- test → starts the sequence
- tb_top → instantiates DUT + interface + starts UVM assign full = (count == DEPTH);

Top Module:

```
module adder(  
    input logic [3:0] a, b,  
    output logic [4:0] sum  
);  
    assign sum = a + b;  
endmodule
```

```
`include "uvm_macros.svh"  
import uvm_pkg::*;
```

// 1. Interface

```
interface adder_if;  
    logic [3:0] a, b;  
    logic [4:0] sum;  
endinterface
```

// 2. Sequence item

```
class adder_seq_item extends uvm_sequence_item;  
    `uvm_object_utils(adder_seq_item)  
    rand logic [3:0] a;  
    rand logic [3:0] b;  
    logic [4:0] sum;  
    function new(string name = "adder_seq_item");  
        super.new(name);  
    endfunction  
    function string convert2string();  
        return $sformatf("a=%0d b=%0d sum=%0d", a, b, sum);  
    endfunction  
endclass
```

// 3. Sequence

```
class adder_sequence extends uvm_sequence #(adder_seq_item);  
    `uvm_object_utils(adder_sequence)  
    function new(string name = "adder_sequence");  
        super.new(name);  
    endfunction  
endclass
```

```

endfunction
task body();
    adder_seq_item item;
    repeat(10) begin
        item = adder_seq_item::type_id::create("item");
        start_item(item);
        assert(item.randomize());
        finish_item(item);
    end
endtask
endclass

```

// 4. Driver

```

class adder_driver extends uvm_driver #(adder_seq_item);
    `uvm_component_utils(adder_driver)
    virtual adder_if vif;
    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction
    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        if(!uvm_config_db #(virtual adder_if)::get(
            this, "", "vif", vif))
            `uvm_fatal("NO_VIF", "adder_if not found")
    endfunction
    task run_phase(uvm_phase phase);
        adder_seq_item item;
        forever begin
            seq_item_port.get_next_item(item);
            vif.a = item.a;
            vif.b = item.b;
            #10;
            `uvm_info("DRV",
                $sformatf("Driving a=%0d b=%0d", item.a, item.b),
                UVM_MEDIUM)
            seq_item_port.item_done();
        end
    endtask
endclass

```

// 5. Monitor

```

class adder_monitor extends uvm_monitor;
    `uvm_component_utils(adder_monitor)
    virtual adder_if vif;
    uvm_analysis_port #(adder_seq_item) ap;
    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction
    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        ap = new("ap", this);
        if(!uvm_config_db #(virtual adder_if)::get(
            this, "", "vif", vif))
            `uvm_fatal("NO_VIF", "adder_if not found")
    endfunction
    task run_phase(uvm_phase phase);
        adder_seq_item item;

```

```

    forever begin
        #10;
        item = adder_seq_item::type_id::create("mon_item");
        item.a = vif.a;
        item.b = vif.b;
        item.sum = vif.sum;
        ap.write(item);
        `uvm_info("MON",
            $sprintf("Observed: %s", item.convert2string()),
            UVM_MEDIUM)
    end
endtask
endclass

```

// 6. Scoreboard

```

class adder_scoreboard extends uvm_scoreboard;
    `uvm_component_utils(adder_scoreboard)
    uvm_analysis_imp #(adder_seq_item, adder_scoreboard) ap;
    int pass_count = 0;
    int fail_count = 0;
    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction
    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        ap = new("ap", this);
    endfunction
    function void write(adder_seq_item item);
        logic [4:0] expected = item.a + item.b;
        if(item.sum === expected) begin
            `uvm_info("SB",
                $sprintf("PASS: %s | expected=%0d",
                    item.convert2string(), expected),
                UVM_MEDIUM)
            pass_count++;
        end
        else begin
            `uvm_error("SB",
                $sprintf("FAIL: %s | expected=%0d got=%0d",
                    item.convert2string(), expected, item.sum))
            fail_count++;
        end
    endfunction
    function void report_phase(uvm_phase phase);
        `uvm_info("SB",
            $sprintf("=== SCOREBOARD: %0d PASS %0d FAIL ===",
                pass_count, fail_count),
            UVM_NONE)
    endfunction
endclass

```

// 7. Agent

```

class adder_agent extends uvm_agent;
    `uvm_component_utils(adder_agent)
    adder_driver          drv;
    adder_monitor          mon;
    uvm_sequencer #(adder_seq_item) seqr;

```

```

function new(string name, uvm_component parent);
    super.new(name, parent);
endfunction
function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    drv = adder_driver::type_id::create("drv", this);
    mon = adder_monitor::type_id::create("mon", this);
    seqr = uvm_sequencer #(adder_seq_item)::type_id::create("seqr", this);
endfunction
function void connect_phase(uvm_phase phase);
    drv.seq_item_port.connect(seqr.seq_item_export);
endfunction
endclass

```

// 8. Environment

```

class adder_env extends uvm_env;
    `uvm_component_utils(adder_env)
    adder_agent agent;
    adder_scoreboard sb;
    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction
    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        agent = adder_agent::type_id::create("agent", this);
        sb = adder_scoreboard::type_id::create("sb", this);
    endfunction
    function void connect_phase(uvm_phase phase);
        agent.mon.ap.connect(sb.ap);
    endfunction
endclass

```

// 9. Test

```

class adder_test extends uvm_test;
    `uvm_component_utils(adder_test)
    adder_env env;
    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction
    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        env = adder_env::type_id::create("env", this);
    endfunction
    task run_phase(uvm_phase phase);
        adder_sequence seq;
        phase.raise_objection(this);
        seq = adder_sequence::type_id::create("seq");
        seq.start(env.agent.seqr);
        phase.drop_objection(this);
    endtask
endclass

```

// 10. Top module

```

module tb_top;
    adder_if aif();

    adder dut(

```

```

        .a(aif.a),
        .b(aif.b),
        .sum(aif.sum)
    );

    initial begin
        uvm_config_db #(virtual adder_if)::set(
            null, "uvm_test_top.*", "vif", aif);
        run_test("adder_test");
    end
endmodule

```

Tools: Cadence Xcelium(Eda playground)

Compile options: -timescale 1ns/1ns -sv

Run options: -access +rw +UVM_VERBOSITY=UVM_MEDIUM +UVM_TESTNAME=adder_test

Output:

```

xrun: 25.03-s001: (c) Copyright 1995-2025 Cadence Design Systems, Inc.
    Top level design units:
        uvm_pkg
        cdns_uvmapi
        cdns_assert2uvm_pkg
        cdns_uvm_pkg
        $unit_0x67f934e9
        tb_top
Loading snapshot worklib.tb_top:sv ..... Done
SVSEED default: 1
xcelium> source /xcelium25.03/tools/xcelium/files/xmsimrc
xcelium> source /xcelium25.03/tools/methodology/UVM/CDNS-1.1d/additions/sv/files/tcl/uvm_sim.tcl
xcelium> run

UVM_INFO @ 0: reporter [RNTST] Running test adder_test...
UVM_INFO testbench.sv(121) @ 10: uvm_test_top.env.sb [SB] PASS: a=8 b=13 sum=21 | expected=21
UVM_INFO testbench.sv(97) @ 10: uvm_test_top.env.agent.mon [MON] Observed: a=8 b=13 sum=21
UVM_INFO testbench.sv(65) @ 10: uvm_test_top.env.agent.drv [DRV] Driving a=8 b=13

UVM_INFO testbench.sv(121) @ 20: uvm_test_top.env.sb [SB] PASS: a=5 b=7 sum=12 | expected=12
UVM_INFO testbench.sv(97) @ 20: uvm_test_top.env.agent.mon [MON] Observed: a=5 b=7 sum=12
UVM_INFO testbench.sv(65) @ 20: uvm_test_top.env.agent.drv [DRV] Driving a=5 b=7

UVM_INFO testbench.sv(121) @ 30: uvm_test_top.env.sb [SB] PASS: a=15 b=13 sum=28 | expected=28
UVM_INFO testbench.sv(97) @ 30: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=13 sum=28
UVM_INFO testbench.sv(65) @ 30: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=13

UVM_INFO testbench.sv(121) @ 40: uvm_test_top.env.sb [SB] PASS: a=7 b=11 sum=18 | expected=18
UVM_INFO testbench.sv(97) @ 40: uvm_test_top.env.agent.mon [MON] Observed: a=7 b=11 sum=18
UVM_INFO testbench.sv(65) @ 40: uvm_test_top.env.agent.drv [DRV] Driving a=7 b=11

UVM_INFO testbench.sv(121) @ 50: uvm_test_top.env.sb [SB] PASS: a=13 b=2 sum=15 | expected=15
UVM_INFO testbench.sv(97) @ 50: uvm_test_top.env.agent.mon [MON] Observed: a=13 b=2 sum=15
UVM_INFO testbench.sv(65) @ 50: uvm_test_top.env.agent.drv [DRV] Driving a=13 b=2

UVM_INFO testbench.sv(121) @ 60: uvm_test_top.env.sb [SB] PASS: a=15 b=2 sum=17 | expected=17
UVM_INFO testbench.sv(97) @ 60: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=2 sum=17
UVM_INFO testbench.sv(65) @ 60: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=2

UVM_INFO testbench.sv(121) @ 70: uvm_test_top.env.sb [SB] PASS: a=15 b=13 sum=28 | expected=28
UVM_INFO testbench.sv(97) @ 70: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=13 sum=28
UVM_INFO testbench.sv(65) @ 70: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=13

UVM_INFO testbench.sv(121) @ 80: uvm_test_top.env.sb [SB] PASS: a=12 b=4 sum=16 | expected=16
UVM_INFO testbench.sv(97) @ 80: uvm_test_top.env.agent.mon [MON] Observed: a=12 b=4 sum=16
UVM_INFO testbench.sv(65) @ 80: uvm_test_top.env.agent.drv [DRV] Driving a=12 b=4

UVM_INFO testbench.sv(121) @ 90: uvm_test_top.env.sb [SB] PASS: a=6 b=5 sum=11 | expected=11
UVM_INFO testbench.sv(97) @ 90: uvm_test_top.env.agent.mon [MON] Observed: a=6 b=5 sum=11
UVM_INFO testbench.sv(65) @ 90: uvm_test_top.env.agent.drv [DRV] Driving a=6 b=5

UVM_INFO testbench.sv(121) @ 100: uvm_test_top.env.sb [SB] PASS: a=14 b=13 sum=27 | expected=27
UVM_INFO testbench.sv(97) @ 100: uvm_test_top.env.agent.mon [MON] Observed: a=14 b=13 sum=27
UVM_INFO testbench.sv(65) @ 100: uvm_test_top.env.agent.drv [DRV] Driving a=14 b=13

```

UVM_INFO /xcelium25.03/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_objection.svh(1268) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase

UVM_INFO testbench.sv(135) @ 100: uvm_test_top.env.sb [SB] === SCOREBOARD: 10 PASS 0 FAIL ===

--- UVM Report catcher Summary ---

Number of demoted UVM_FATAL reports : 0
Number of demoted UVM_ERROR reports : 0
Number of demoted UVM_WARNING reports: 0
Number of caught UVM_FATAL reports : 0
Number of caught UVM_ERROR reports : 0
Number of caught UVM_WARNING reports : 0

--- UVM Report Summary ---

** Report counts by severity

UVM_INFO : 33
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0

** Report counts by id

[DRV] 10
[MON] 10
[RNTST] 1
[SB] 11
[TEST_DONE] 1

Simulation complete via \$finish(1) at time 100 NS + 49

/xcelium25.03/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_root.svh:457 \$finish;
xcelium> exit